



Date: 27 / 12 / 2025

Lab Practical #4:

Implementation of monoalphabetic and polyalphabetic substitution cipher technique.

Program:

```
const A = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
const B = "ZYXWVUTSRQPONMLKJIHGFEDCBA";

function enc(s: string): string {
  return s.split('').map(c => {
    const i = A.indexOf(c.toUpperCase());
    return i !== -1
      ? (c === c.toUpperCase() ? B[i] : B[i].toLowerCase())
      : c;
  }).join('');
}

function dec(s: string): string {
  return enc(s);
}

const t = "Hello World";
console.log(`Text: ${t}`);
console.log(`Encrypted: ${enc(t)}`);
console.log(`Decrypted: ${dec(enc(t))}`);
```

Output:

```
~/Desktop/sem-6/ins main*
) bun monoalphabetic-cipher.ts
Text: Hello World
Encrypted: Svool Dliow
Decrypted: Hello World
```



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Program:

```
function enc(s: string, k: string): string {
    let r = "";
    let ki = 0;
    for (const c of s) {
        if (/[A-Za-z]/.test(c)) {
            const sh = k.toUpperCase().charCodeAt(ki % k.length) - 65;
            const b = c.toUpperCase().charCodeAt(0) - 65;
            const x = (b + sh) % 26;
            r += c === c.toUpperCase()
                ? String.fromCharCode(x + 65)
                : String.fromCharCode(x + 65).toLowerCase();
            ki++;
        } else r += c;
    }
    return r;
}

function dec(s: string, k: string): string {
    let r = "";
    let ki = 0;
    for (const c of s) {
        if (/[A-Za-z]/.test(c)) {
            const sh = k.toUpperCase().charCodeAt(ki % k.length) - 65;
            const b = c.toUpperCase().charCodeAt(0) - 65;
            const x = (b - sh + 26) % 26;
            r += c === c.toUpperCase()
                ? String.fromCharCode(x + 65)
                : String.fromCharCode(x + 65).toLowerCase();
            ki++;
        } else {
            r += c;
        }
    }
    return r;
}

const t = "Hello World";
const k = "KEY";
console.log(`Text: ${t}`);
console.log(`Key: ${k}`);
console.log(`Encrypted: ${enc(t, k)}`);
console.log(`Decrypted: ${dec(enc(t, k), k)}`);
```



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Output:

```
~/Desktop/sem-6/ins main*  
> bun polyalphabetic-cipher.ts  
Text: Hello World  
Key: KEY  
Encrypted: Rijvs Uyvjn  
Decrypted: Hello World
```